# From Perception to Punchline:
## Empowering VLM with the Art of In-the-wild Memes

# HUMOR

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Hierarchical Understanding and Meme Optimization

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## 01 What makes a caption a meme?

**Same visual relation, different context-specific punchlines.**

**Template** **Meme**




**USER INPUT**
**Technology · "Apple fanatic"**

| A · Subject | B · New focus | C · Neglected |
|---|---|---|
| APPLE FANS | IPHONE 11 | IPHONE 10 |

### Why rank within a group?

Single-image scores barely separate quality. In-group ranks make differences easier to compare.

## 02 HUMOR: hierarchical reasoning → local preferences → stable optimization

### A Hierarchical CoT

Template intent → diverse contextual groundings.

**1 MULTI-SCENARIO ASSOCIATION**



→ A confusing work problem → bewildered reaction

→ A bad joke → exaggerated "painful" response

→ Social ignorance → refusing to listen

Trace back to the compatible scene



**2 REVERSE-ANCHOR FROM A REAL MEME**
Gold c → infer U: socializing / ignorance.
Trace back; retain other associations.

$$\tilde{r} = \arg\max_r P_\phi(c \mid r, I, U)$$

$$\mathbb{E}_{r\sim P_\theta}[\tilde{h}_G(r)] \geq 1 - (1-\alpha)\delta$$

Good-path mass α > 0; mean humor gap ≤ δ.

**INFERENCE Explore/select; no gold caption.**

### B Template-local reward

Compare within a template; learn a group ranking.

Training pairs span five types: wrong text, wrong location, boring, detailed boring, and generated memes.

**Original** ≻ **Boring**







**Preference** $p_{ij}^G = \sigma[s_\phi(m_i) - s_\phi(m_j)]$

**BCE loss** $\ell = -y\log p - (1-y)\log(1-p)$

**Expected wins** $\mathrm{EBC}_G(i) = \sum_{j\neq i} p_{ij}^G$

1 Estimate preferences inside template G.
2 Sum wins; normalize sparse comparisons.
3 Rank candidates → local policy targets.

### C Group-wise optimization

Use the group ranking as the policy target. Stay close to the CoT reference during optimization.

1 Sample captions from the CoT policy
2 Pairwise reward → EBC → local targets
3 Update policy; keep CoT as reference

$$\mathcal{L} = -\mathbb{E}[\sum_k q_G(m_k)\log \pi_\theta(c_k \mid I)] + \beta\Delta_{\mathrm{KL}}$$

$$\Delta_{\mathrm{KL}} = \mathbb{E}_I[\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{CoT}})]$$

$$H(\pi_\theta) \geq H(\pi_{\mathrm{CoT}}) - \sqrt{\Delta_{\mathrm{KL}}/2}$$

For h ∈ [0,1], KL bounds possible degradation. It does not guarantee improvement.

## 03 Does HUMOR improve meme quality and diversity?

| Model / variant | Hum. ↑ | Read. ↑ | Rel. ↑ | Orig. ↑ | Sim. ↑ | Dist. ↑ | Human % ↑ |
|---|---|---|---|---|---|---|---|
| Qwen2.5-7B | 2.39 | 3.35 | 2.91 | 2.57 | 0.576 | 0.564 | 75.7 |
| Qwen2.5-32B | 2.54 | 3.52 | 3.09 | 2.76 | 0.564 | 0.566 | 82.2 |
| InternVL3-8B | 2.39 | 2.79 | 3.04 | 2.79 | 0.545 | 0.564 | 62.7 |
| GLM-4.1V-9B-Thinking | 1.73 | 2.62 | 2.75 | 2.71 | 0.602 | 0.572 | 45.1 |
| Keye-VL-8B-preview | 2.35 | 3.19 | 2.99 | 2.71 | 0.585 | 0.580 | 69.0 |
| GPT-4o | 2.70 | 2.99 | 3.21 | **2.97** | 0.603 | 0.552 | 91.3 |
| Gemini-2.5-flash | 2.81 | 3.29 | 3.25 | 2.88 | 0.600 | 0.561 | — |
| **HUMOR-CoT** | 2.68 | **3.70** | 3.50 | 2.90 | **0.640** | 0.590 | 91.5 |
| CoT: Single Path | 1.87 | 2.79 | 2.68 | 2.45 | 0.637 | 0.570 | 86.0 |
| CoT: Self-Improve | 2.38 | 3.68 | 3.00 | 2.65 | 0.629 | 0.578 | 89.1 |
| CoT: Subquestion | 1.85 | 3.32 | 2.58 | 2.47 | 0.639 | **0.597** | 87.2 |
| **HUMOR-RL (preview)** | **2.83** | 3.67 | **3.55** | 2.79 | 0.631 | 0.588 | **92.3** |

Human scores: 30 Prolific raters, 0–5. Human Rate: Gemini-2.5-pro.
Sim.: bge-base-en-v1.5. Dist.: context-swapped generations. Bold = best.

**CoT vs. 7B base**

| | |
|---|---|
| Humor | +0.29 |
| Readability | +0.35 |
| Relevance | +0.59 |
| Originality | +0.33 |
| Similarity | +0.064 |
| Context swap | +0.026 |
| Human Rate | +15.8 pp |

**RL preview: 2.83 humor**
Orig. 2.90 → 2.79

## 04 Meme generation across themes

| | Human reference | Single-path CoT | HUMOR-CoT |
|---|---|---|---|
| **Christmas**<br>Love · humor<br>Family tradition<br>Delayed excitement |  |  |  |
| **Romance**<br>Happiness · humor<br>Crush · dabbing<br>Rare moment |  |  |  |



## 05 Does the reward recover human preference rankings?

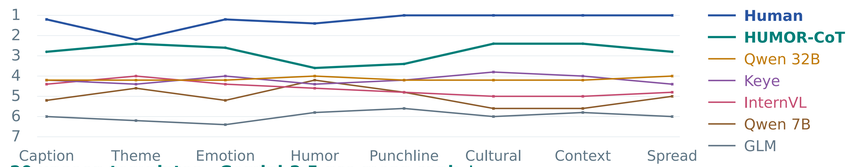
| Reward backbone / state | G1 · τ ↑ | G2 · τ ↑ | G3 · τ ↑ | G4 · τ ↑ | G5 · τ ↑ |
|---|---|---|---|---|---|
| Qwen2.5-VL-7B · base | 0.16 | 0.28 | 0.47 | -0.10 | 0.29 |
| Qwen2.5-VL-7B · trained | 0.47 | 0.56* | 0.42 | 0.14 | 0.47 |
| Qwen2.5-VL-32B · base | 0.16 | 0.16 | -0.02 | 0.14 | 0.29 |
| Qwen2.5-VL-32B · trained | 0.29 | 0.47* | 0.07 | 0.30 | 0.42 |
| Keye-VL-8B · base | 0.05 | 0.09 | 0.16 | 0.29 | 0.16 |
| **Keye-VL-8B · trained** | **0.78*** ** | **0.77*** ** | **0.78*** ** | **0.78*** ** | **0.78*** ** |

100 MaxDiff raters · 5 templates · *p<.05, ***p<.001

**Human MaxDiff ranks vs. model pairwise + EBC ranks.**

**Keye aligns best; Qwen gains are mixed or limited.**

## 06 Does HUMOR generalize to unseen templates?




**20 unseen templates · Gemini-2.5-pro mean rank ↓**



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